Q4.

4.1. We can see by visual inspection that:

$$c_3 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = c_1 + c_2 \Rightarrow \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

and:. if ~~a vector ca~~ $c \Rightarrow \lambda a + \mu b$. then c is a linear comb. of a & b. $\therefore$ c_3 is l.c. of c_1 & c_2.

4.2. Basis for column space of A $\Rightarrow \{c_1, c_2\} \Rightarrow \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 10 \end{bmatrix} \right\}$.

The drone can move in the region covered by the vectors formed by l.c. of c_1 & c_2.

~~We can see that~~

$$\text{lc of } c_1, c_2 \rightarrow \lambda c_1 + \mu c_2 \Rightarrow (\lambda+\mu)\, i + \lambda j + \mu k.$$

$\therefore$ lc of c_1, c_2 spans the plane $\boxed{x = y + z}$

$\therefore$ Drone's movement is restricted in this plane.

4.3. No. this vector is not inside Column Space of A.

as $\boxed{0 \neq 0 + 5.}$